Querying RMBS trustee data in natural language: results

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A language model converts each question into a structured query spec, and a compiler turns the spec into SQL using metric definitions held in code, so the model never does the arithmetic. These results come from a working prototype on synthetic data: 50 Australian RMBS deals, 21,945 loans and 773,336 loan-months, in 2 trustee formats.

Test results

test	result
Cleaned table against simulated truth	0 differences across 773,336 loan-months
Resubmitted duplicates removed	721 of 774,057 raw rows
Clean-up calls reported as repayments, recovered	7 deals
Compiler output against hand-written SQL	49 of 49 match, worst difference 1.6e-15
Compiled queries valid for Oracle	49 of 49
Slowest question at 27,840,096 loan-months	under 0.75 s, default threads

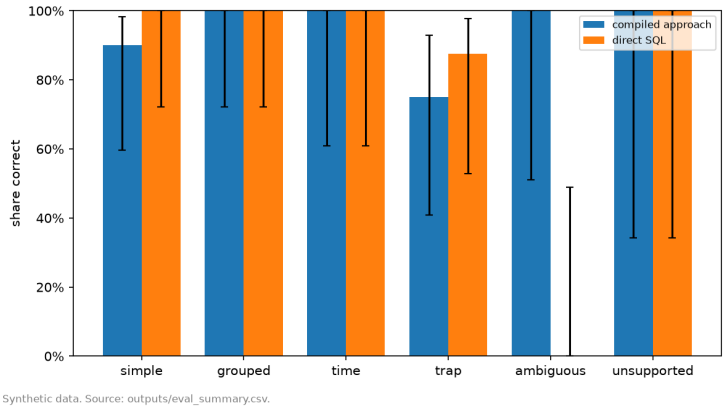
What common mistakes do to the numbers

mistake	question	correct	with the mistake
2 missed payments read as 60+ days	Q27	0.66%	0.59%
Clean-up calls counted as prepayments	Q29	21.6%	100.0%
Clean-up calls counted as prepayments	Q30	18.3%	49.0%
Exit month counted as a live loan	Q32	9,975	10,572
Simple average, not balance-weighted	Q33	51.96%	50.92%
Monthly CPR averaged, not compounded (dropped)	Q2 2025 CPR	18.064%	18.063%

Compared with direct SQL

Same 40 questions, with the same table and metric definitions in the direct SQL model's prompt. Neither approach returned an incorrect number; every miss on both sides was a refusal to answer. On ambiguous questions, the compiled approach asked what was meant, where direct SQL declined without explanation.

category	questions	compiled approach	direct SQL
simple	10	9 (90%)	10 (100%)
grouped	10	10 (100%)	10 (100%)
time	6	6 (100%)	6 (100%)
trap	8	6 (75%)	7 (88%)
ambiguous	4	4 (100%)	0 (0%)
unsupported	2	2 (100%)	2 (100%)
total	40	37 (92%)	35 (88%)



Synthetic data. Source: outputs/eval_summary.csv.

Asked "What are arrears in NSW?", the compiled approach replied:

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Which arrears measure do you want for NSW?
Options: 30+ days past due; 90+ days past due
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